

# [HELPER FOR NSUER]

Weekly Progress Report

---



**Department of Electrical and Computer Engineering  
North South University  
CSE 299: Junior Project Design**

**Group Number- 1**

**Week – 2**

|             |                          |
|-------------|--------------------------|
| <b>Name</b> | <b>ISRAT ZAMAN SRITY</b> |
| <b>ID</b>   | <b>2211084042</b>        |

# 1. PROGRESS & ACHIEVEMENTS

## • 1. PROGRESS & ACHIEVEMENTS

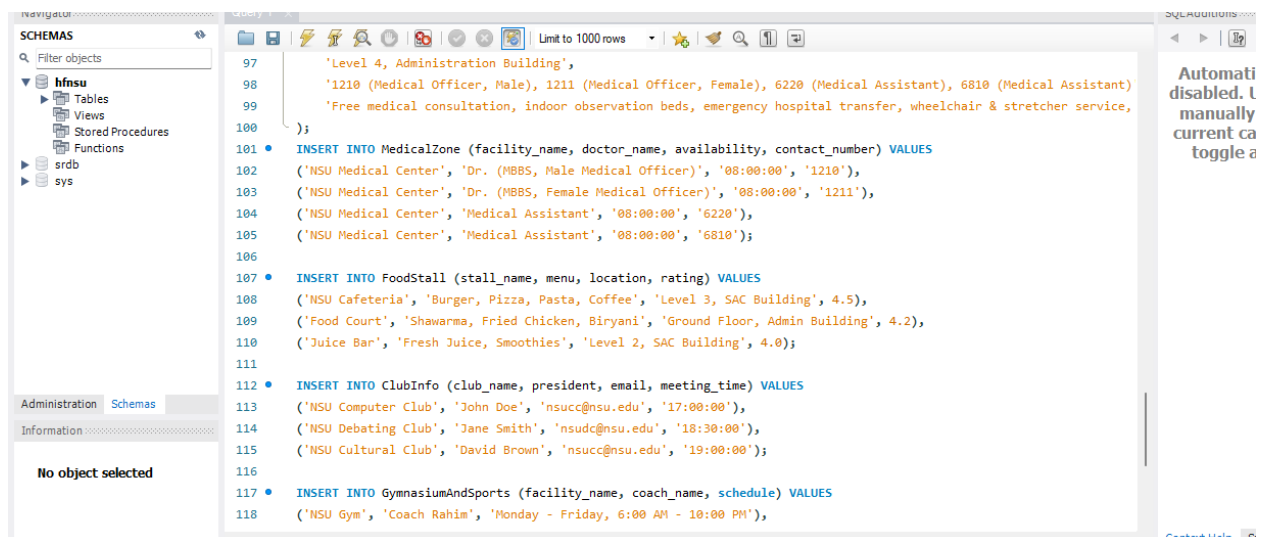
This week, significant progress was made in designing and implementing the database structure for the project. The work primarily involved defining and setting up tables, inserting relevant data, and ensuring that the database aligns with the project requirements. Below are the detailed contributions and achievements:

### Database Schema Design & Implementation

- Developed a relational database schema that includes key tables such as Library, MedicalZone, Foodstall, ClubInfo, Gym and sports, Administration, Parking and Printzone .
- Ensured data normalization to prevent redundancy and improve query efficiency.

### Data Insertion & Population

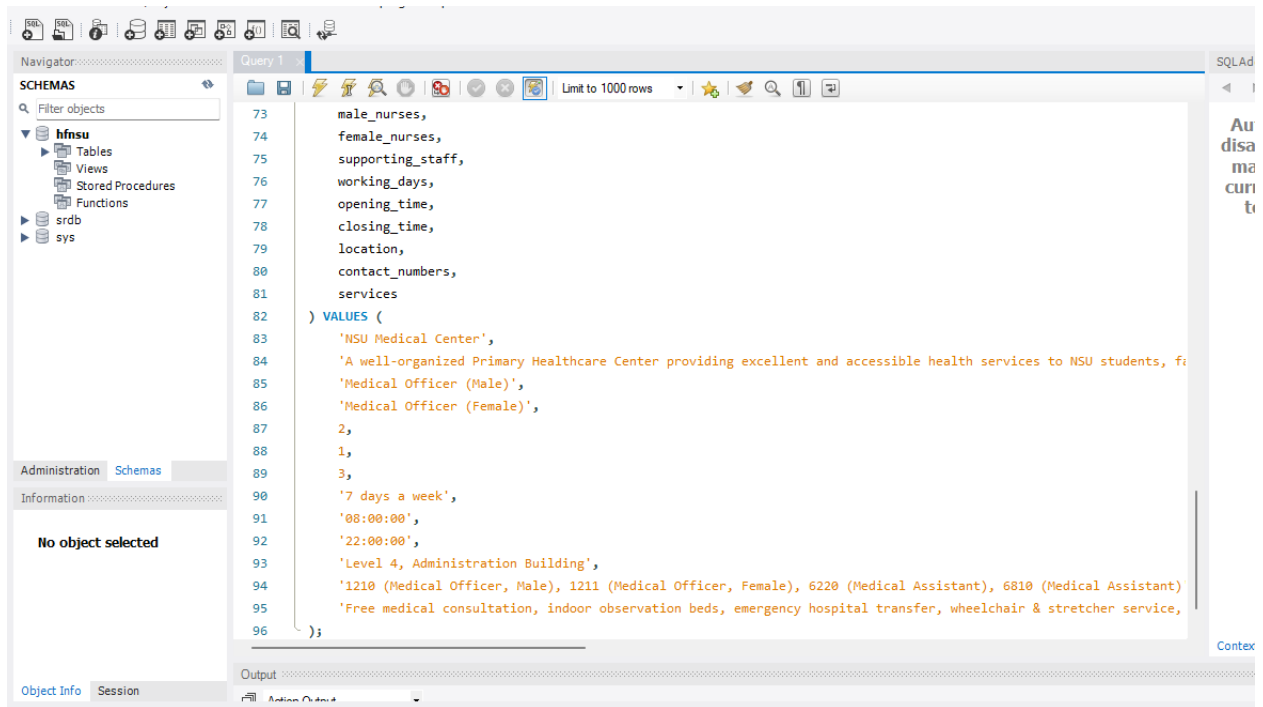
- Inserted initial data into each table, ensuring relevance and accuracy.



The screenshot displays a SQL IDE interface. On the left, a 'SCHEMAS' pane shows a tree view with 'hfnisu' expanded, containing 'Tables', 'Views', 'Stored Procedures', 'Functions', 'srdb', and 'sys'. The main editor shows SQL queries for inserting data into 'MedicalZone', 'FoodStall', 'ClubInfo', and 'GymnasiumAndSports'. The 'MedicalZone' query includes details for 'Level 4, Administration Building' and lists medical officers and assistants. The 'FoodStall' query lists items like 'Burger, Pizza, Pasta, Coffee' and 'Shawarma, Fried Chicken, Biryani'. The 'ClubInfo' query lists clubs like 'NSU Computer Club' and 'NSU Debating Club'. The 'GymnasiumAndSports' query lists 'NSU Gym' and 'Coach Rahim'. A right-hand pane shows a message: 'Automation disabled. I manually current ca toggle a'. The bottom status bar indicates 'Context Help'.

```
97 'Level 4, Administration Building',
98 '1210 (Medical Officer, Male), 1211 (Medical Officer, Female), 6220 (Medical Assistant), 6810 (Medical Assistant)
99 'Free medical consultation, indoor observation beds, emergency hospital transfer, wheelchair & stretcher service,
100 ');
101 • INSERT INTO MedicalZone (facility_name, doctor_name, availability, contact_number) VALUES
102 ('NSU Medical Center', 'Dr. (MBBS, Male Medical Officer)', '08:00:00', '1210'),
103 ('NSU Medical Center', 'Dr. (MBBS, Female Medical Officer)', '08:00:00', '1211'),
104 ('NSU Medical Center', 'Medical Assistant', '08:00:00', '6220'),
105 ('NSU Medical Center', 'Medical Assistant', '08:00:00', '6810');
106
107 • INSERT INTO FoodStall (stall_name, menu, location, rating) VALUES
108 ('NSU Cafeteria', 'Burger, Pizza, Pasta, Coffee', 'Level 3, SAC Building', 4.5),
109 ('Food Court', 'Shawarma, Fried Chicken, Biryani', 'Ground Floor, Admin Building', 4.2),
110 ('Juice Bar', 'Fresh Juice, Smoothies', 'Level 2, SAC Building', 4.0);
111
112 • INSERT INTO ClubInfo (club_name, president, email, meeting_time) VALUES
113 ('NSU Computer Club', 'John Doe', 'nsucc@nsu.edu', '17:00:00'),
114 ('NSU Debating Club', 'Jane Smith', 'nsudc@nsu.edu', '18:30:00'),
115 ('NSU Cultural Club', 'David Brown', 'nsucc@nsu.edu', '19:00:00');
116
117 • INSERT INTO GymnasiumAndSports (facility_name, coach_name, schedule) VALUES
118 ('NSU Gym', 'Coach Rahim', 'Monday - Friday, 6:00 AM - 10:00 PM'),
```

- Created structured queries to insert data into MedicalZone, including doctors, nurses, and healthcare facilities available.



- Populated the Library table with books, authors, and ISBN numbers to simulate a real-world university library database.
- Updated the FoodStall table with stall names, menus, locations, and customer ratings.
- Collected a substantial amount of data for various tables; however, many of them have not been inserted yet due to challenges in organizing them effectively.

### Query Testing & Debugging

- Tested INSERT, SELECT, UPDATE, and DELETE queries to ensure smooth database operation.

Navigation pane on the left shows a database schema with tables: hnsu, srdb, sys. The main editor displays SQL code for inserting data into MedicalZone, FoodStall, ClubInfo, and GymnasiumAndSports tables. The code includes facility names, doctor/coach names, availability, contact numbers, menus, locations, and ratings. A sidebar on the right contains a message: "Automati disabled. L manually current ca toggle a".

The screenshot shows a SQL editor with a query to select all data from the Library table. Below the editor is a "Result Grid" showing the results of the query. The grid has columns: id, book\_name, author, isbn, and available\_copies. The results show three rows of data. Below the result grid is an "Action Output" pane showing the execution of the query, including the time taken and the number of rows affected.

| id | book_name                                  | author               | isbn          | available_copies |
|----|--------------------------------------------|----------------------|---------------|------------------|
| 1  | Introduction to Algorithms                 | Thomas H. Cormen     | 9780262033848 | 5                |
| 2  | Database System Concepts                   | Abraham Silberschatz | 9780078022159 | 3                |
| 3  | Artificial Intelligence: A Modern Approach | Stuart Russell       | 9780134610993 | 4                |

Action Output:

| # | Time     | Action                                                                                     | Message                                                |
|---|----------|--------------------------------------------------------------------------------------------|--------------------------------------------------------|
| 3 | 01:05:55 | INSERT INTO Library (book_name, author, isbn, available_copies) VALUES (Introduction to... | 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0 |
| 4 | 01:06:07 | select * from Library LIMIT 0, 1000                                                        | 3 row(s) returned                                      |

The screenshot shows a SQL editor with a query to set the rating for the FoodStall table. The query sets the rating to 4.7 for the stall named 'Healthy Bites'. Below the editor is a "Result Grid" showing the results of the query. The grid has columns: id, stall\_name, menu, location, and rating. The results show three rows of data. Below the result grid is an "Action Output" pane showing the execution of the query, including the time taken and the number of rows affected.

| id | stall_name    | menu                             | location                     | rating |
|----|---------------|----------------------------------|------------------------------|--------|
| 1  | NSU Cafeteria | Burger, Pizza, Pasta, Coffee     | Level 3, SAC Building        | 4.5    |
| 2  | Food Court    | Shawarma, Fried Chicken, Biryani | Ground Floor, Admin Building | 4.2    |
| 3  | Juice Bar     | Fresh Juice, Smoothies           | Level 2, SAC Building        | 4.0    |

Action Output:

| # | Time | Action | Message |
|---|------|--------|---------|
|---|------|--------|---------|

## Concepts Learning

- Deepened understanding of **database normalization** and best practices for structuring large-scale data.
  - Enhanced knowledge of **query optimization**, ensuring that data retrieval is efficient.
  - Learning effective ways to handle **TEXT data types** for storing long descriptive content.
- 

## • 2. OBSTACLES

Despite the progress made, several challenges were encountered during the development phase. These obstacles, along with the mitigation strategies, are discussed below:

### 1. Data Compatibility Issues

- **Issue:** Some values, especially long descriptions in the MedicalZone and SuggestionBox tables, exceeded standard VARCHAR limits.
- **Mitigation:** Used TEXT and LONGTEXT data types instead of VARCHAR where necessary.

### 2. Data Insertion Constraints

- **Issue:** Certain tables, such as MedicalZone, required multiple entries for different personnel but had constraints that needed adjusting.
- **Mitigation:** Trying to modified table constraints to allow multiple medical officers and assistants while maintaining data integrity.

### 3. External Dependencies & Delays

- **Issue:** Gathering structured data, especially for clubs, administration, and medical zones, required validation from multiple sources.
- **Mitigation:** Still working on it , couldn't concluded up what to do yet .

### 4. Data Organization Issues

- **Issue:** A significant amount of data has been collected, but due to challenges in structuring and organizing it, much of it has not been inserted into the database yet.

- **Mitigation:** Trying to developing a systematic approach to categorize and organize the collected data before insertion to ensure accuracy and consistency.

## 2. MILESTONES & REPORTING

Describe the planned timeline and any updates to the original plan here. You can show the changes to the original plan in different colour. (This part can be same for all group members)

| Milestone | Tasks                                    | Date       |
|-----------|------------------------------------------|------------|
| Week-1    | Project proposal.                        | 29/01/2025 |
| Week-2    | <del>Project proposal presentation</del> | 5/02/2025  |
| Week-3    | Sign up/ Log in Page creation.           | 19/02/2025 |
| Week-4    | Create database                          | 26/02/2025 |
| Week-5    | Proposal Report                          |            |
| Week-6    | UI/UX completion.                        |            |
| Week-7    | database set up.                         |            |
| Week-8    | Progress Report                          |            |
| Week-9    | Front end complete.                      |            |
| Week-10   | Back end complete.                       |            |
| Week-11   | Deployment                               |            |
| Week-12   | Documentation                            |            |
| Week-13   | Final Presentation AND Report            |            |